1) Najděte $\vec{u} = \overrightarrow{AB}$, nakreslete 3 umístění vektoru $\vec{u}$, najděte X tak, aby $\overrightarrow{CX} = \vec{u}$

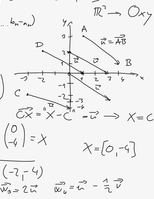
$A = [1,3]$ $B = [4,1]$ $C = [-3,-2]$

$\mathbb{R}^2 \to Oxy$

$\overrightarrow{AB} = "B - A" = (b_1 - a_1, b_2 - a_2, \dots b_n - a_n)$
$= (3, -2) = \vec{u}$

umístění $\vec{u}$ do $D = [-2,2]$

$D + \vec{u} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$



X tak, aby $\overrightarrow{CX} = \vec{u}$

$\overrightarrow{CX} = "X - C" = \vec{u} \to X = C + \vec{u}$

$C + \vec{u} = \begin{pmatrix} -3 \\ -2 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \end{pmatrix} = X$

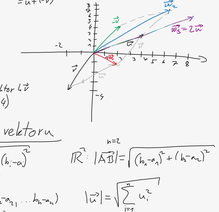
$X = [0, -4]$

2) $\vec{u} = (4,3)$ $\vec{v} = (-2,-4)$

$\vec{w_1} = \vec{u} + \vec{v}$ $\vec{w_2} = \vec{u} - \vec{v} = \vec{u} + (-\vec{v})$ $\vec{w_3} = 2\vec{u}$ $\vec{w_4} = \vec{u} - \frac{1}{2}\vec{v}$

$\vec{w_1} = (2,-1)$
$\vec{w_2} = (6,7)$
$\vec{w_3} = (8,6)$
$\vec{w_4} = (5,5)$

$-\vec{v}$ je opačný vektor k $\vec{v}$
$-\vec{v} = -1 \cdot \vec{v} = (2,4)$



Velikost vektoru

$|\overrightarrow{AB}| = \sqrt{\sum_{i=1}^{n} (b_i - a_i)^2}$

$\mathbb{R}^2$ $|\overrightarrow{AB}| = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2}$

$\overrightarrow{AB} = (b_1 - a_1, b_2 - a_2, \dots b_n - a_n)$
$\vec{u} = (u_1, u_2, \dots u_n)$

$\mathbb{R}^n$ $|\vec{u}| = \sqrt{\sum_{i=1}^{n} u_i^2}$

$\mathbb{R}^2$ $|\vec{u}| = \sqrt{u_1^2 + u_2^2}$

$\mathbb{R}^3$ $|\vec{u}| = \sqrt{u_1^2 + u_2^2 + u_3^2}$

Př

$\vec{u} = (1,1)$

$|\vec{u}| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\vec{v} = (0,-3,2) \in \mathbb{R}^3$ $|\vec{v}| = \sqrt{0^2 + (-3)^2 + 2^2} = \sqrt{13}$

Najděte vektor rovnoběžný s $\vec{u}$ tak, aby $|\vec{v}| = 1$

$\vec{u} = (1,3,-2)$ $|\vec{u}| = \sqrt{1 + 9 + 4} = \sqrt{14}$

$\vec{v} = \alpha \cdot \vec{u}$ $\alpha \in \mathbb{R} \to \alpha = \frac{1}{|\vec{u}|} = \frac{1}{\sqrt{14}}$

$|\vec{v}| = 1$

$\to \vec{v} = \frac{1}{\sqrt{14}} (1, 3, -2) = \left(\frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}}, \frac{-2}{\sqrt{14}}\right)$

$|\vec{v}| = \sqrt{\left(\frac{1}{\sqrt{14}}\right)^2 + \left(\frac{3}{\sqrt{14}}\right)^2 + \left(\frac{-2}{\sqrt{14}}\right)^2} = \sqrt{\frac{1}{14} + \frac{9}{14} + \frac{4}{14}} = \sqrt{\frac{14}{14}} = 1$

"normalizace"

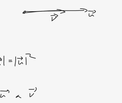
Skalární součin

$\vec{u} \cdot \vec{v} = \sum_{i=1}^{n} u_i \cdot v_i = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$

$|\vec{u}| = \sqrt{\sum_{i=1}^{n} u_i^2} = \sqrt{\sum_{i=1}^{n} u_i \cdot u_i} = \sqrt{\vec{u} \cdot \vec{u}} \iff |\vec{u}|^2 = \vec{u} \cdot \vec{u}$

Skalární součin a úhel

$\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos\varphi$



$\cos\varphi = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|} = \left(\frac{\vec{u}}{|\vec{u}|}\right) \cdot \left(\frac{\vec{v}}{|\vec{v}|}\right)$

$|\ |=1$ $|\ |=1$

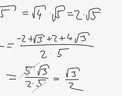
$\varphi = 0 \to \cos\varphi = 1$

$\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}|$ $\vec{u} \cdot \vec{u} = |\vec{u}| \cdot |\vec{u}| = |\vec{u}|^2$

Určete úhel mezi vektory $\vec{u}$ a $\vec{v}$

$\vec{u} = (1,1)$ $|\vec{u}| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\vec{v} = (2,0)$ $|\vec{v}| = \sqrt{2^2 + 0^2} = \sqrt{4} = 2$

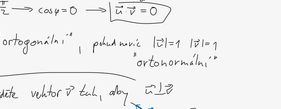


$\cos\varphi = \frac{1 \cdot 2 + 1 \cdot 0}{\sqrt{2} \cdot 2} = \frac{2}{\sqrt{2} \cdot 2} = \frac{\sqrt{2}}{2}$

$\cos\varphi = \frac{\sqrt{2}}{2}$

$\boxed{\varphi_1 = \frac{\pi}{4}}$

$\varphi_2 = \frac{7}{4}\pi$



$\varphi = ?$ $\vec{u} = (-1,2)$ $\vec{v} = (-\sqrt{3}+2, 1+2\sqrt{3})$

$|\vec{u}| = \sqrt{1^2 + 2^2} = \sqrt{5}$

$|\vec{v}| = \sqrt{(2-\sqrt{3})^2 + (1+2\sqrt{3})^2} = \sqrt{4 - 4\sqrt{3} + 3 + 1 + 4\sqrt{3} + 4 \cdot 3}$

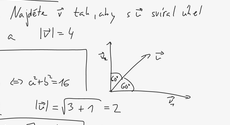
$= \sqrt{20} = \sqrt{4 \cdot 5} = \sqrt{4} \cdot \sqrt{5} = 2 \cdot \sqrt{5}$

$\cos\varphi = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|} = \frac{-1(2-\sqrt{3}) + 2(1+2\sqrt{3})}{\sqrt{5} \cdot 2\sqrt{5}} = \frac{-2+\sqrt{3}+2+4\sqrt{3}}{2 \cdot 5}$

$= \frac{5\sqrt{3}}{2 \cdot 5} = \frac{\sqrt{3}}{2}$

$\cos\varphi = \frac{\sqrt{3}}{2}$

$\varphi = 30°$
$= \frac{\pi}{6}$



$\vec{u} \perp \vec{v}$

$\varphi = \frac{\pi}{2} \to \cos\varphi = 0 \to \boxed{\vec{u} \cdot \vec{v} = 0}$

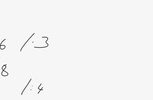
"ortogonální", pokud navíc $|\vec{u}| = 1$ $|\vec{v}| = 1$ "ortonormální"

Najděte vektor $\vec{v}$ tak, aby $\vec{u} \perp \vec{v}$

$\vec{u} = (2,1)$

$\vec{v_1} = (-1,2)$ $\vec{u} \cdot \vec{v} = 2 \cdot (-1) + 1 \cdot 2 = 0$

$\vec{v_2} = (1,-2)$



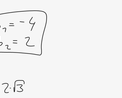
· $\vec{u} = (\sqrt{3}, -1)$ Najděte $\vec{v}$ tak, aby s $\vec{u}$ svíral úhel $\varphi = 60°$ a $|\vec{v}| = 4$

$\vec{v} = (a,b)$

$|\vec{v}| = \sqrt{a^2 + b^2} = 4 \iff a^2 + b^2 = 16$

$\frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|} = \cos 60° = \frac{1}{2}$

$|\vec{u}| = \sqrt{3 + 1} = 2$



$\boxed{\begin{aligned} \frac{1}{2} &= \frac{\sqrt{3}a - b}{2 \cdot 4} \quad (1) \\ a^2 + b^2 &= 16 \quad (2) \end{aligned}}$

(1) $4 = \sqrt{3}a - b$

$a = \frac{1}{\sqrt{3}}(b+4)$

(2) $\left(\frac{1}{\sqrt{3}}(b+4)\right)^2 + b^2 = 16$

$\frac{1}{3}(b^2 + 8b + 16) + b^2 = 16 \quad / \cdot 3$

$b^2 + 8b + 16 + 3b^2 = 48$

$4b^2 + 8b - 32 = 0 \quad / : 4$

$b^2 + 2b - 8 = 0$

$(b+4)(b-2) = 0$

$\boxed{\begin{aligned} b_1 &= -4 \\ b_2 &= 2 \end{aligned}}$

$a = \frac{1}{\sqrt{3}}(b+4)$ $a_1 = \frac{1}{\sqrt{3}}(-4+4) = 0$

$a_2 = \frac{1}{\sqrt{3}}(2+4) = \frac{6}{\sqrt{3}} = 2\sqrt{3}$

$\vec{v} = (a,b)$ $\vec{v_1} = (0,-4)$ $|\vec{v_1}| = 4$

$\vec{v_2} = (2\sqrt{3}, 2)$ $|\vec{v_2}| = \sqrt{12 + 4} = 4$

Lineární kombinace

$\vec{v} = \alpha \vec{u_1} + \beta \vec{u_2}$

Vektor $\vec{z}$ napište jako LK vektorů $\vec{u}$ a $\vec{v}$

$\vec{z} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $\vec{u} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ $\vec{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$\vec{z} = x \cdot \vec{u} + y \cdot \vec{v}$, $x, y \in \mathbb{R}$

$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = x \begin{pmatrix} -3 \\ 4 \end{pmatrix} + y \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3x + 2y \\ 4x + y \end{pmatrix}$

$\left.\begin{aligned} -3x + 2y &= 1 \\ 4x + y &= 1 \quad / \cdot (-2) \end{aligned}\right\} \oplus$

$-11x + 0y = -1$

$x = \frac{1}{11}$

$-\frac{3}{11} + 2y = 1$

$y = \frac{7}{11}$

$\frac{1}{11} \underbrace{\begin{pmatrix} -3 \\ 4 \end{pmatrix}}_{\vec{u}} + \frac{7}{11} \underbrace{\begin{pmatrix} 2 \\ 1 \end{pmatrix}}_{\vec{v}} =$

$= \begin{pmatrix} \frac{-3}{11} \\ \frac{4}{11} \end{pmatrix} + \begin{pmatrix} \frac{14}{11} \\ \frac{7}{11} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \vec{z}$ ✓

$\vec{z} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ $\vec{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\vec{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\vec{z} = 2\vec{u} + 3 \cdot \vec{v}$